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The Survivor's Number

Problem 0.6 of the Mathematical Problem Solving projects

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Abstract

People numbered 1 to n stand in a circle. Starting with person 1, every second person is eliminated until only one is left, and $J(n)$ denotes the number of this survivor. We first played the game by hand for $n = 1, \ldots, 16$. The results suggested three rules: $J(2k) = 2J(k) - 1$, $J(2k+1) = 2J(k) + 1$, and the fact that every n can be written in exactly one way as $n = 2^m + \ell$ with $0 \leq \ell < 2^m$. We proved them with one lemma: the game only depends on the places of the people around the circle, not on their numbers, so after one turn of the circle it becomes a smaller copy of itself. We then proved the formula

$$J(2^m + \ell) = 2\ell + 1$$

in two ways, by induction on m and by induction on n. We generalised it to a game that starts with any person s: the survivor is the person 2ℓ places after s. In base 2, the formula becomes a simple rule, since $J(n)$ is obtained by moving the first digit of n to the end. This gives at once, for example, $J(10^9) = 926\,258\,177$. Finally, we explain the Python code we wrote to simulate the game and to check every result.

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1 Introduction

This report presents the work of Group 6 on Problem 0.6 of the project sheet of the course Mathematical Problem Solving [2, 1], entitled *Survivor's number*.

1.1 The problem

The statement given in the project sheet [2] is the following.

> We start with n people numbered 1 to n around a circle, and we eliminate every second remaining person until only one survives. For example, for $n = 10$, the elimination order is $2, 4, 6, 8, 10$, then $3, 7, 1, 9$, so 5 survives. Determine the survivor's number, $J(n)$. For example, $J(10) = 5$. The case $n = 0$ makes no sense. First, check that
>
> $$J(1) = 1, \quad J(2) = 1, \quad J(3) = 3, \quad J(4) = 1, \quad J(5) = 3, \quad J(6) = 5.$$

We make the rules precise.

Definition 1 (The elimination game) *Let $n \geq 1$. We place n people, numbered $1, 2, \ldots, n$, around a circle, and after person n comes person 1 again. We go around the circle, always in the same direction, and apply the following rules:*

- *we skip one person, then we eliminate the next one, then we skip the next one, and so on;*
- *an eliminated person leaves the circle, so only the people still present are counted;*
- *the game stops when only one person is left.*

The number of the last person is the survivor's number.

This elimination game is a classical problem, treated for example in [3, Chapter 1].

In the example of the statement, with $n = 10$, the game skips 1 and eliminates 2, skips 3 and eliminates 4, and so on until 10. It then comes back to the beginning of the circle and continues with the people still present, until only person 5 is left.

1.2 Why we need a formula

For ten people, the game can be played on paper in a minute. With 1000 people it needs 999 eliminations, and after the first turn the people left are no longer consecutive numbers, so it becomes hard to follow who comes next. So the question is: *can we find $J(n)$ without playing the whole game?* Our objectives are to

1. find a formula that gives $J(n)$ directly for every $n \geq 1$, and prove it;

2. find the survivor when the game starts with any person;

3. find the survivor quickly for very large circles, and check our results with a computer program.

1.3 Who starts the game

In the statement, the first person eliminated is person 2. So person 1 is the first person skipped: the game starts with person 1. Until the end of Section 4.1 we keep this assumption, and $J(n)$ always means the survivor of this game. This is not a restriction: a game that starts with another person is the same game turned around the circle. Once $J(n)$ is known, we use this remark in Section 4.2 to treat any starting person.

1.4 Method and plan of the report

We followed the method of the course [1]: play with small cases, look for a pattern, make a precise guess, and prove it so that it holds for every n and not only for the cases we tested. The report follows the same order.

- **Section 2** plays the game for $n = 1, \ldots, 16$ and collects the results in a table.
- **Section 3** proves a key lemma, which says that the game only depends on the places of the people around the circle. With it, we prove how the survivor for $2k$ and $2k + 1$ people depends on the survivor for k people, and we show that every n is written in one way as a power of 2 plus a remainder.
- **Section 4** proves the formula for $J(n)$ in two ways, by induction on m and by induction on n, gives its consequences, and generalises it to a game that starts with any person.
- **Section 5** writes the formula in base 2, which gives the survivor at once for very large circles.
- **Section 6** explains the Python code we wrote to play the game and to check our results.
- **Section 7** concludes.

2 Playing the Game by Hand

Before looking for a formula, we play the game by hand for every n from 1 to 16, starting with person 1 as in Section 1.3. The results are collected in Table 1 at the end of this section.

2.1 Conventions

A *round* is one turn of the circle. It ends when the game comes back to the beginning of the circle, and the next round goes on with the people still present. When n is odd, the first round skips person n and eliminates person 1, who comes right after him: we count this elimination in the first round.

Up to $n = 7$, and for $n = 16$, we draw the circle at the end of each round and write under it the people eliminated during that round, in order. From $n = 8$ to $n = 15$ we draw only the final circle, with the rounds written under it. The symbols are the following:



2.2 The sixteen tests

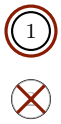
2.2.1 For $n = 1, 2, 3$

$n = 1$



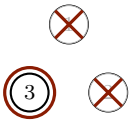
No elimination
$J(1) = 1$

$n = 2$



Round 1: 2
$J(2) = 1$

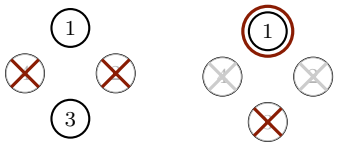
$n = 3$



Round 1: 2, 1
$J(3) = 3$

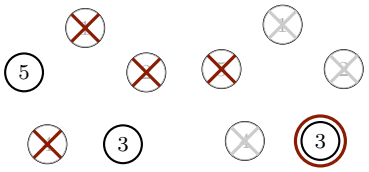
2.2.2 For $n = 4$ to 7

$n = 4$



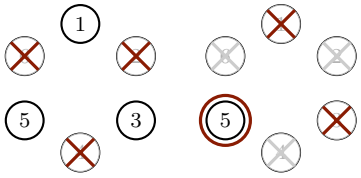
Round 1: 2, 4 | Round 2: 3
$J(4) = 1$

$n = 5$



Round 1: 2, 4, 1 | Round 2: 5
$J(5) = 3$

$n = 6$

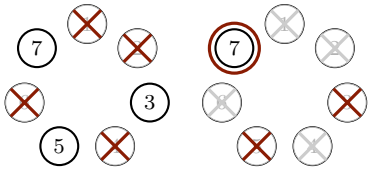


Round 1: 2, 4, 6

Round 2: 3, 1

$J(6) = 5$

$n = 7$



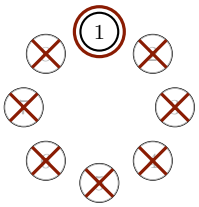
Round 1: 2, 4, 6, 1

Round 2: 5, 3

$J(7) = 7$

2.2.3 For $n = 8$ to 15

$n = 8$



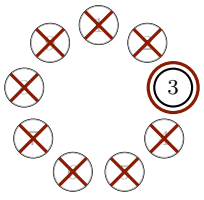
Round 1: 2, 4, 6, 8

Round 2: 3, 7

Round 3: 5

$J(8) = 1$

$n = 9$



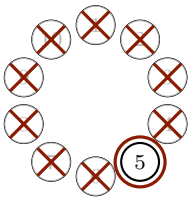
Round 1: 2, 4, 6, 8, 1

Round 2: 5, 9

Round 3: 7

$J(9) = 3$

$n = 10$



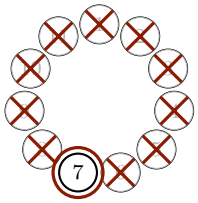
Round 1: 2, 4, 6, 8, 10

Round 2: 3, 7, 1

Round 3: 9

$J(10) = 5$

$n = 11$



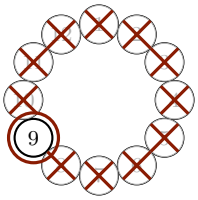
Round 1: 2, 4, 6, 8, 10, 1

Round 2: 5, 9, 3

Round 3: 11

$J(11) = 7$

$n = 12$



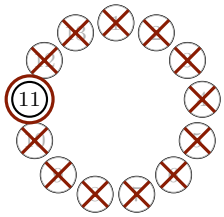
Round 1: 2, 4, 6, 8, 10, 12

Round 2: 3, 7, 11

Round 3: 5, 1

$J(12) = 9$

$n = 13$



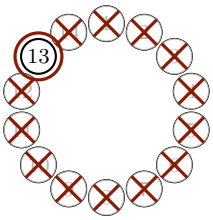
Round 1: 2, 4, 6, 8, 10, 12, 1

Round 2: 5, 9, 13

Round 3: 7, 3

$J(13) = 11$

$n = 14$



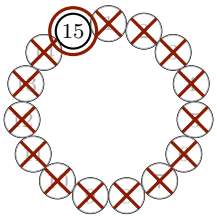
Round 1: 2, 4, 6, 8, 10, 12, 14

Round 2: 3, 7, 11, 1

Round 3: 9, 5

$J(14) = 13$

$n = 15$



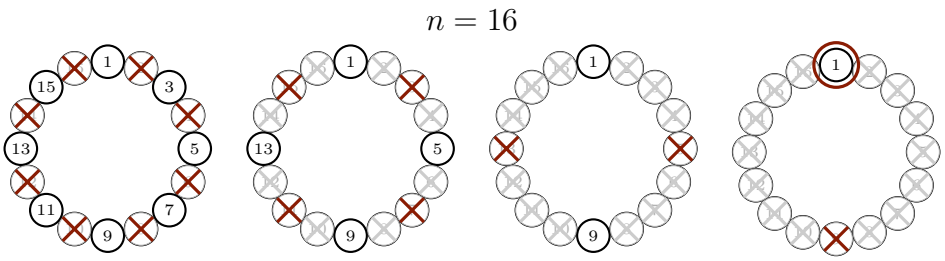
Round 1: 2, 4, 6, 8, 10, 12, 14, 1

Round 2: 5, 9, 13, 3

Round 3: 11, 7

$J(15) = 15$

2.2.4 For $n = 16$



Round 1: 2, 4, 6, 8, 10, 12, 14, 16 | Round 2: 3, 7, 11, 15 | Round 3: 5, 13 | Round 4: 9 $J(16) = 1$

2.3 Summary of the tests

We write each n as the largest power of 2 that is not bigger than n, plus a remainder: $n = 2^m + \ell$ with $0 \leq \ell < 2^m$. Table 1 puts this decomposition next to the survivor found in the tests.

n	$n = 2^m + \ell$	m	ℓ	$J(n)$	$2\ell + 1$
1	$2^0 + 0$	0	0	1	1
2	$2^1 + 0$	1	0	1	1
3	$2^1 + 1$	1	1	3	3
4	$2^2 + 0$	2	0	1	1
5	$2^2 + 1$	2	1	3	3
6	$2^2 + 2$	2	2	5	5
7	$2^2 + 3$	2	3	7	7
8	$2^3 + 0$	3	0	1	1
9	$2^3 + 1$	3	1	3	3
10	$2^3 + 2$	3	2	5	5
11	$2^3 + 3$	3	3	7	7
12	$2^3 + 4$	3	4	9	9
13	$2^3 + 5$	3	5	11	11
14	$2^3 + 6$	3	6	13	13
15	$2^3 + 7$	3	7	15	15
16	$2^4 + 0$	4	0	1	1

Table 1: The tests for $n = 1, \ldots, 16$. The horizontal lines separate the blocks $2^m \leq n < 2^{m+1}$.

Three things stand out in Table 1.

- All the survivors are odd.
- The survivor is person 1 when $n = 1, 2, 4, 8, 16$, that is when n is a power of 2.
- Inside each block between two lines, the survivor goes up by 2 each time a person is added. In every line, $J(n)$ is equal to $2\ell + 1$.

These are only remarks on sixteen cases. We prove them in Sections 3 and 4.

3 What the Table Shows, and Why

Table 1 suggests several rules. In this section we state them one by one and prove each of them, so that they hold for every number of people and not only for the sixteen circles we played. We first prove a lemma that we use in all the proofs: the game only depends on the places of the people, not on their numbers. The first two rules then link a circle of $2k$ or $2k+1$ people to a circle of k people. The third one is about the way each n is written in the table.

3.1 The key lemma: only the places matter

At any moment of the game, the people still present stand around the circle in a fixed order. We write them as a list, starting with the person who is skipped next.

Definition 2 *A* list $(a_1, a_2, \ldots, a_k)$ *is the sequence of the k people still present, in the order of the circle, where a_1 is the next person to be skipped. If $k \geq 2$, one* step *of the game skips a_1 and eliminates a_2. The next person to be skipped is then a_3, and a_1 is now the last person before we come back to a_3:*

$$(a_1, a_2, a_3, \ldots, a_k) \longrightarrow (a_3, a_4, \ldots, a_k, a_1).$$

For $k = 2$, the new list is (a_1). When $k = 1$ the game is over and a_1 is the survivor.

Each step removes one person, so a game with k people ends after $k-1$ steps. The game of Definition 1 starts from the list $(1, 2, \ldots, n)$. For example, with $n = 5$:

$$(1, 2, 3, 4, 5) \to (3, 4, 5, 1) \to (5, 1, 3) \to (3, 5) \to (3).$$

The people eliminated are $2, 4, 1, 5$, in this order, and the survivor is 3, as in Table 1.

The step rule only uses the places in the list, never the numbers of the people. This is what the next lemma says.

Lemma 1 (Only the places matter) *Let $(a_1, \ldots, a_k)$ be a list of $k \geq 1$ people. The survivor of the game that starts from this list is $a_{J(k)}$, the person at place $J(k)$.*

Proof. We prove by induction on $k \geq 1$ the statement

$$L(k): \quad \text{for every list } (a_1, \ldots, a_k) \text{ of } k \text{ people, the survivor is } a_{J(k)}.$$

Base case. For $k = 1$, the only person a_1 is the survivor, and $J(1) = 1$. So $L(1)$ is true.

Induction step. Let $k \geq 2$ and assume $L(k-1)$. Take a list $(a_1, \ldots, a_k)$. One step gives the list of $k-1$ people

$$(b_1, \ldots, b_{k-1}) = (a_3, a_4, \ldots, a_k, a_1),$$

that is, $b_j = a_{f(j)}$ with

$$f(j) = \begin{cases} j+2 & \text{if } 1 \le j \le k-2, \\ 1 & \text{if } j = k-1. \end{cases}$$

The game goes on from this new list, so by $L(k-1)$ its survivor is $b_{J(k-1)} = a_{f(J(k-1))}$. This is true for every list. For the list $(1, 2, \ldots, k)$, which is the usual game with k people, the survivor is $J(k)$ by definition, so $J(k) = f\big(J(k-1)\big)$. Going back to any list, the survivor is therefore $a_{f(J(k-1))} = a_{J(k)}$, and $L(k)$ is true.

By induction, $L(k)$ is true for every $k \ge 1$. ■

The proof also gives a relation between $J(k)$ and $J(k-1)$. Since $J(k) = f\big(J(k-1)\big)$,

$$J(k) = \begin{cases} J(k-1)+2 & \text{if } J(k-1) \le k-2, \\ 1 & \text{if } J(k-1) = k-1. \end{cases} \tag{1}$$

In words: adding one person moves the survivor two places, except when the survivor was the last person, in which case it goes back to person 1. This is what we saw in Table 1. Relation (1) gives $J(n)$ for every n, but it needs $n-1$ steps. The two relations of the next subsections are much faster, because they divide n by 2.

3.2 Even circles: $J(2k) = 2J(k) - 1$

3.2.1 What we observe

We put side by side the survivor for k people and the survivor for $2k$ people.

k	1	2	3	4	5	6	7	8
$J(k)$	1	1	3	1	3	5	7	1
$J(2k)$	1	1	5	1	5	9	13	1
$2J(k)-1$	1	1	5	1	5	9	13	1

The last two lines are equal in all eight cases.

Observation 1 *For every $k \ge 1$, $J(2k) = 2J(k) - 1$.*

3.2.2 Proof

Let $k \ge 1$, and let $n = 2k$ people stand around the circle. The idea is to play the first round completely, and to see that what is left is a game with k people.

Step 1: the first round eliminates exactly the even numbers. We show by induction on j, for $j = 1, \ldots, k$, the statement

$E(j)$: the first j eliminations skip $1, 3, \ldots, 2j-1$ and eliminate $2, 4, \ldots, 2j$.

Base case. The game starts by skipping person 1 and eliminating person 2, so $E(1)$ is true.

Induction step. Let $1 \leq j < k$ and assume $E(j)$. At this point the people eliminated are $2, 4, \ldots, 2j$, and the last one is $2j$. The next person is $2j+1$. The people $2j+1$ and $2j+2$ are still present, and $2j+2 \leq 2k$. So the game skips $2j+1$ and eliminates $2j+2$, and $E(j+1)$ is true.

By $E(k)$, the k-th person eliminated is $2k$, the last person of the circle. The next person is person 1, who is still present: we are back at the beginning of the circle and the first round is over. It has eliminated $2, 4, \ldots, 2k$, and the k people left are, in the order of the circle,

$$1,\ 3,\ 5,\ \ldots,\ 2k-1.$$

The game goes on by skipping person 1.

Step 2: the people left form a game with k people. We write the people left as a list $(a_1, a_2, \ldots, a_k)$, in the order of the circle, starting with the person who is skipped next:

$$a_i = 2i - 1 \qquad (i = 1, \ldots, k),$$

so $a_1 = 1$, $a_2 = 3$, $\ldots$, $a_k = 2k-1$ (Figure 1). The rest of the game is the game of Definition 1 played on $a_1, \ldots, a_k$ instead of $1, \ldots, k$: the circle has k people and it starts by skipping the first one.

Step 3: the survivor is $a_{J(k)}$. By Lemma 1, applied to the list $(a_1, \ldots, a_k)$, the survivor is the person at place $J(k)$ of this list. So

$$J(2k) = a_{J(k)} = 2J(k) - 1.$$

This proves Observation 1 for every $k \geq 1$. □

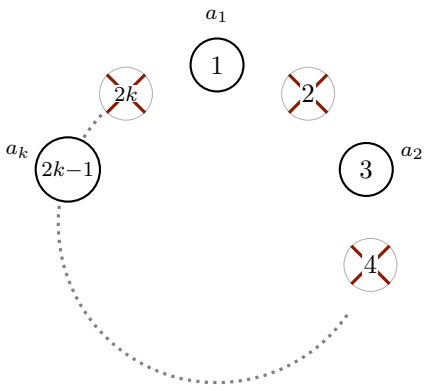


Figure 1: The circle of $2k$ people after the first round: $a_i = 2i - 1$. The dotted arc stands for the people not drawn.

Example. For $n = 10$, the first round leaves $(1, 3, 5, 7, 9)$. Since $J(5) = 3$, the survivor is the third person of this list, $a_3 = 5$. So $J(10) = 2 \times 3 - 1 = 5$, as in Table 1.

3.3 Odd circles: $J(2k+1) = 2J(k)+1$

3.3.1 What we observe

We now compare the survivor for k people and for $2k+1$ people.

k	1	2	3	4	5	6	7
$J(k)$	1	1	3	1	3	5	7
$J(2k+1)$	3	3	7	3	7	11	15
$2J(k)+1$	3	3	7	3	7	11	15

Again, the last two lines are equal in all seven cases.

Observation 2 *For every $k \geq 1$, $J(2k+1) = 2J(k)+1$.*

3.3.2 Proof

Let $k \geq 1$, and let $n = 2k+1$ people stand around the circle. We follow the same three steps as for $2k$ people. Only the end of the first round changes.

Step 1: the first round. The induction of the previous proof works without change: the first k eliminations skip $1, 3, \ldots, 2k-1$ and eliminate $2, 4, \ldots, 2k$. Indeed, for $j < k$ we still have $2j+2 \leq 2k < 2k+1$.

After the k-th elimination (person $2k$), one person remains before the end of the circle: person $2k+1$. He is skipped. The next person is person 1, since after person $2k+1$ we come back to the beginning of the circle, and person 1 is eliminated. As said in Section 2.1, this elimination belongs to the first round. So the first round eliminates $2, 4, \ldots, 2k$ and then 1, and the k people left are

$$3,\ 5,\ 7,\ \ldots,\ 2k+1.$$

The game goes on with the first person after 1 who is still present. Person 2 is already eliminated, so this is person 3, and he is the next one skipped.

Step 2: the people left form a game with k people. This time the list starts with person 3:

$$a_i = 2i+1 \qquad (i = 1, \ldots, k),$$

so $a_1 = 3$, $a_2 = 5$, $\ldots$, $a_k = 2k+1$ (Figure 2). The rest of the game is again the game of Definition 1 with k people, played on $a_1, \ldots, a_k$.

Step 3: the survivor. By Lemma 1, the survivor is the person at place $J(k)$ of the list, so

$$J(2k+1) = a_{J(k)} = 2J(k)+1.$$

This proves Observation 2 for every $k \geq 1$. □

Example. For $n = 11$, the first round eliminates $2, 4, 6, 8, 10$ and then 1, and leaves $(3, 5, 7, 9, 11)$. Since $J(5) = 3$, the survivor is $a_3 = 7$. So $J(11) = 2\times 3+1 = 7$, as in Table 1.

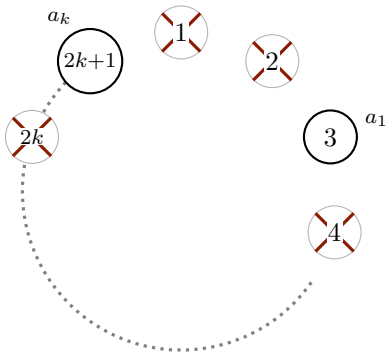


Figure 2: The circle of $2k+1$ people after the first round: $a_i = 2i+1$. Person 1 is eliminated at the end of the round, so the list starts with person 3.

3.3.3 Computing $J(n)$ by halving

With $J(1) = 1$, Observations 1 and 2 give $J(n)$ for every n. We divide n by 2 (rounding down) until we reach 1, then we go back up with the two relations. For $n = 100$:

$$100 \to 50 \to 25 \to 12 \to 6 \to 3 \to 1,$$

and

$$J(3) = 2J(1)+1 = 3, \qquad J(6) = 2J(3)-1 = 5, \qquad J(12) = 2J(6)-1 = 9,$$
$$J(25) = 2J(12)+1 = 19, \quad J(50) = 2J(25)-1 = 37, \quad J(100) = 2J(50)-1 = 73.$$

For $n = 1000$, the halvings are $1000 \to 500 \to 250 \to 125 \to 62 \to 31 \to 15 \to 7 \to 3 \to 1$, and going back up gives

$$J(3) = 3, \qquad J(7) = 7, \qquad J(15) = 15, \qquad J(31) = 31, \qquad J(62) = 61,$$
$$J(125) = 123, \quad J(250) = 245, \quad J(500) = 489, \quad J(1000) = 977.$$

About $\log_2 n$ steps replace the $n-1$ eliminations of the game: 6 steps for $n = 100$ and 9 steps for $n = 1000$. Still, this does not tell us what $J(n)$ looks like. For that we go back to the way the numbers are written in Table 1.

3.4 Every n is $2^m + \ell$ in exactly one way

3.4.1 What we observe

In Table 1, each n is written as $2^m + \ell$, where 2^m is the largest power of 2 that is not bigger than n:

$$5 = 2^2 + 1, \qquad 10 = 2^3 + 2, \qquad 15 = 2^3 + 7, \qquad 16 = 2^4 + 0.$$

In every line the remainder ℓ is smaller than 2^m, and there is only one such way to write n. For instance, $10 = 2^2 + 6$ is not allowed, because $6 \geq 2^2$.

Observation 3 *Every integer $n \geq 1$ can be written in exactly one way as*

$$n = 2^m + \ell, \qquad m \geq 0, \quad 0 \leq \ell < 2^m.$$

3.4.2 Proof by induction

Existence, by strong induction on n. For $n = 1$ we have $1 = 2^0 + 0$, and $0 < 2^0$.

Let $n \geq 2$, and assume that every integer k with $1 \leq k < n$ can be written $k = 2^p + r$ with $0 \leq r < 2^p$. We write $n = 2k$ if n is even, or $n = 2k + 1$ if n is odd. In both cases $1 \leq k < n$, so $k = 2^p + r$ with $0 \leq r < 2^p$.

- If $n = 2k$, then $n = 2^{p+1} + 2r$, and $0 \leq 2r < 2 \cdot 2^p = 2^{p+1}$.
- If $n = 2k + 1$, then $n = 2^{p+1} + (2r + 1)$. Since $r \leq 2^p - 1$, we get $2r + 1 \leq 2^{p+1} - 1 < 2^{p+1}$.

In both cases $n = 2^m + \ell$ with $m = p + 1$ and $0 \leq \ell < 2^m$.

Uniqueness. The condition $0 \leq \ell < 2^m$, with $\ell = n - 2^m$, means

$$2^m \leq n < 2^{m+1}, \tag{2}$$

since $\ell \geq 0$ means $2^m \leq n$, and $\ell < 2^m$ means $n < 2^m + 2^m = 2^{m+1}$. Assume that $n = 2^m + \ell = 2^{m'} + \ell'$ are two such writings, with $m < m'$. Then $m + 1 \leq m'$, and (2) for m and for m' gives

$$n < 2^{m+1} \leq 2^{m'} \leq n,$$

which is impossible. In the same way $m' < m$ is impossible. So $m = m'$, and then $\ell = n - 2^m = \ell'$. □

The proof also shows how to find m and ℓ: by (2), 2^m is the largest power of 2 not bigger than n, and $\ell = n - 2^m$. For example, $64 \leq 100 < 128$ gives $100 = 2^6 + 36$. The integers with the same m form the blocks $2^m \leq n < 2^{m+1}$ separated by lines in Table 1.

The three results of this section are the tools we need. In the next section we use them to prove a formula for $J(n)$.

4 The Formula and Its Generalisation

In this section we prove the formula suggested by our tests, using the results of Section 3. We give two proofs, one by induction on m and one by induction on n. Up to now the game always started with person 1. In the second part of this section we drop this assumption and find the survivor when the game starts with any person.

4.1 The formula $J(n) = 2\ell + 1$

4.1.1 What we observe

In every line of Table 1, the column $J(n)$ is equal to the column $2\ell + 1$. In other words, the survivor depends only on the remainder ℓ.

Theorem 1 *Let $n \geq 1$, written as $n = 2^m + \ell$ with $0 \leq \ell < 2^m$. Then*

$$J(n) = 2\ell + 1.$$

4.1.2 Proof by induction on m

For $m \geq 0$, let $P(m)$ be the statement

$$P(m): \quad \text{for every integer } \ell \text{ with } 0 \leq \ell < 2^m, \quad J(2^m + \ell) = 2\ell + 1.$$

By Observation 3, every $n \geq 1$ is equal to $2^m + \ell$ for one pair (m, ℓ) with $0 \leq \ell < 2^m$. So if $P(m)$ is true for every $m \geq 0$, the theorem is proved.

Base case: $m = 0$**.** Then $0 \leq \ell < 2^0 = 1$, so $\ell = 0$, and

$$J(2^0 + 0) = J(1) = 1 = 2 \times 0 + 1.$$

So $P(0)$ is true.

Induction step. Let $m \geq 1$, and assume that $P(m-1)$ is true. Let ℓ be an integer with $0 \leq \ell < 2^m$. We show that $J(2^m + \ell) = 2\ell + 1$, in two cases.

Case 1: ℓ is even, $\ell = 2k$. From $0 \leq 2k < 2^m$ we get $0 \leq k < 2^{m-1}$, and

$$2^m + \ell = 2^m + 2k = 2\left(2^{m-1} + k\right).$$

By Observation 1, then by $P(m-1)$ applied to $2^{m-1} + k$,

$$J(2^m + \ell) = 2J\left(2^{m-1} + k\right) - 1 = 2(2k+1) - 1 = 4k + 1 = 2\ell + 1.$$

Case 2: ℓ is odd, $\ell = 2k + 1$. From $0 \leq 2k + 1 < 2^m$ we get $2k + 1 \leq 2^m - 1$, so $k \leq 2^{m-1} - 1$, that is $0 \leq k < 2^{m-1}$. Moreover

$$2^m + \ell = 2^m + 2k + 1 = 2\left(2^{m-1} + k\right) + 1.$$

By Observation 2, then by $P(m-1)$ applied to $2^{m-1} + k$,

$$J(2^m + \ell) = 2J\left(2^{m-1} + k\right) + 1 = 2(2k+1) + 1 = 4k + 3 = 2\ell + 1.$$

In both cases $J(2^m + \ell) = 2\ell + 1$, so $P(m)$ is true.

Conclusion. $P(0)$ is true, and $P(m-1)$ implies $P(m)$ for every $m \geq 1$. By induction, $P(m)$ is true for every $m \geq 0$: for every $n \geq 1$, written $n = 2^m + \ell$ with $0 \leq \ell < 2^m$, we have $J(n) = 2\ell + 1$. □

Examples. $100 = 2^6 + 36$ gives $J(100) = 73$, the value found by halving in Section 3.3.3. With $2026 = 2^{10} + 1002$, the formula gives $J(2026) = 2 \times 1002 + 1 = 2005$ at once.

Since $\ell = n - 2^m$, the formula can also be written without ℓ:

$$J(n) = 2n - 2^{m+1} + 1,$$

where 2^m is the largest power of 2 not bigger than n. For $n = 10$: $J(10) = 20 - 16 + 1 = 5$.

4.1.3 A second proof, by induction on n

Relation (1) of Section 3.1, which adds one person at a time, gives a second proof of Theorem 1. We prove $J(n) = 2\ell + 1$ by induction on n. For $n = 1 = 2^0 + 0$ we have $J(1) = 1$.

Let $n \geq 2$, and assume the formula for $n-1$: if $n-1 = 2^{m'} + \ell'$ with $0 \leq \ell' < 2^{m'}$, then $J(n-1) = 2\ell' + 1$. There are two cases.

- If $\ell' \leq 2^{m'} - 2$, then $n = 2^{m'} + (\ell' + 1)$ with $\ell' + 1 < 2^{m'}$, so $\ell = \ell' + 1$. Moreover $J(n-1) = 2\ell' + 1 \leq 2^{m'} + \ell' - 1 = n - 2$, so by (1) $J(n) = J(n-1) + 2 = 2\ell' + 3 = 2\ell + 1$.
- If $\ell' = 2^{m'} - 1$, then $n = 2^{m'+1}$, so $\ell = 0$. Here $J(n-1) = 2^{m'+1} - 1 = n - 1$, the last person, so by (1) $J(n) = 1 = 2 \times 0 + 1$.

In both cases $J(n) = 2\ell + 1$. □

The first proof uses the halving of the circle (n becomes about $n/2$), the second one adds one person at a time ($n - 1$ becomes n). They lead to the same formula, which is a good check.

4.1.4 What the formula explains

The formula proves the remarks we made on Table 1.

Corollary 1 *Let $n \geq 1$, written as $n = 2^m + \ell$ with $0 \leq \ell < 2^m$.*

(a) $J(n)$ is odd, and $1 \leq J(n) \leq n$.

(b) $J(n) = 1$ if and only if n is a power of 2.

(c) $J(n) = n$ if and only if $n = 2^{m+1} - 1$, as for $n = 1, 3, 7, 15$.

(d) $J(n+1) = J(n) + 2$ if $n+1$ is not a power of 2, and $J(n+1) = 1$ if $n+1$ is a power of 2.

Proof. By Theorem 1, $J(n) = 2\ell + 1$. (a) The number $2\ell + 1$ is odd and at least 1. Moreover $2\ell + 1 \leq 2^m + \ell$ is the same as $\ell \leq 2^m - 1$, which is true. (b) $2\ell + 1 = 1$ if and only if $\ell = 0$, that is $n = 2^m$. (c) $2\ell + 1 = 2^m + \ell$ if and only if $\ell = 2^m - 1$, that is $n = 2^{m+1} - 1$. (d) If $n + 1$ is not a power of 2, then $\ell + 1 < 2^m$ (otherwise $n + 1 = 2^{m+1}$), so $n + 1 = 2^m + (\ell + 1)$ with the same m, and $J(n + 1) = 2(\ell + 1) + 1 = J(n) + 2$. If $n + 1$ is a power of 2, then $J(n + 1) = 1$ by (b). ■

Point (a) agrees with Step 1 of Section 3.2: every even number leaves during the first round. Point (d) is relation (1) again. The table below summarises these points. On each block $2^m \leq n < 2^{m+1}$, the survivor starts again at 1 and goes through the odd numbers in increasing order, up to the last person $2^{m+1} - 1$.

m	block of n	values of $J(n)$, in order
0	1	1
1	$2, 3$	$1, 3$
2	$4, \ldots, 7$	$1, 3, 5, 7$
3	$8, \ldots, 15$	$1, 3, 5, \ldots, 15$
4	$16, \ldots, 31$	$1, 3, 5, \ldots, 31$
m	$2^m, \ldots, 2^{m+1} - 1$	$1, 3, 5, \ldots, 2^{m+1} - 1$

A block has 2^m values of n, and there are exactly 2^m odd numbers between 1 and $2^{m+1} - 1$, so each odd number appears once in each block.

4.2 When the game starts with person s

So far, person 1 was always the first person skipped (Section 1.3). In a real game we may not know who starts. So we ask: *if the game starts with person s, who survives?*

Definition 3 *Let $n \geq 1$ and $1 \leq s \leq n$. We write $J_s(n)$ for the survivor of the game on the circle $1, 2, \ldots, n$ that starts with person s: we skip s, eliminate the next person, and go on with the rules of Definition 1. In particular $J_1(n) = J(n)$.*

4.2.1 An example: $n = 10$ and $s = 4$

The rounds are now counted from person 4.

- Round 1 skips 4 and eliminates $5, 7, 9, 1, 3$. The game is back at person 4.
- Round 2 skips 4 and eliminates 6, skips 8 and eliminates 10, skips 2 and eliminates 4.
- Only 8 and 2 are left. Round 3 skips 8 and eliminates 2.

The survivor is 8, so $J_4(10) = 8$ (Figure 3).

When person 1 starts, the survivor is $J(10) = 5$. Here the game starts three places later, and the survivor is also three places later: $5 + 3 = 8$.

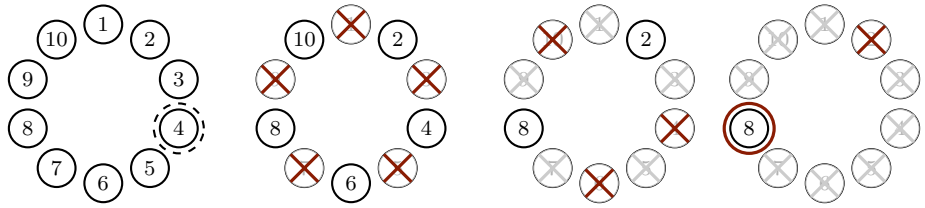


Figure 3: The game with 10 people that starts with person 4 (dashed): $J_4(10) = 8$.

4.2.2 Counting places from the first person

The rule *skip one, eliminate the next* only uses the order of the people around the circle. So we number the *places* from the person who starts: place 1 is person s, place 2 is the person after him, and so on. For $n = 10$ and $s = 4$:

place	1	2	3	4	5	6	7	8	9	10
person	4	5	6	7	8	9	10	1	2	3

By Lemma 1, the survivor is at place $J(10) = 5$, that is person 8. In general the survivor is at place $J(n)$, and we only need to know who stands there. Going forward from s we meet $s+1$, $s+2$, ..., up to person n, and then we come back to person 1. So the person at place i is $s+i-1$ if $s+i-1 \le n$, and $s+i-1-n$ otherwise. For instance, place 8 gives $4+8-1 = 11 > 10$, so person $11-10 = 1$.

4.2.3 The general formula

Since the survivor is at place $J(n) = 2\ell + 1$, we obtain the following result.

Theorem 2 *Let $n \ge 1$ and $1 \le s \le n$, and write $n = 2^m + \ell$ with $0 \le \ell < 2^m$. If the game starts with person s, the survivor is*

$$J_s(n) = \begin{cases} s + 2\ell & \text{if } s + 2\ell \le n, \\ s + 2\ell - n & \text{if } s + 2\ell > n. \end{cases}$$

In words: the survivor is the person 2ℓ places after s around the circle.

Proof. The game that starts with s starts from the list $(a_1, \ldots, a_n) = (s, s+1, \ldots, n, 1, \ldots, s-1)$ of the people in the order of the circle, where a_i is the person at place i. By Lemma 1, the survivor is $a_{J(n)}$, the person at place $J(n)$, and $J(n) = 2\ell+1$ by Theorem 1. The person at place $2\ell+1$ is $s+2\ell$ if $s+2\ell \le n$, and $s+2\ell-n$ otherwise.

It remains to check that one subtraction of n is always enough, that is $s+2\ell-n \le n$. Since $\ell < 2^m$, we have $2\ell < 2^m + \ell = n$, and with $s \le n$ we get $s + 2\ell < 2n$. ■

Another way to read the formula. Since $J(n) = 2\ell + 1$, we have $s + 2\ell = J(n) + (s-1)$. So

$$J_s(n) = J(n) + (s-1), \quad \text{minus } n \text{ if this is bigger than } n.$$

Starting the game $s-1$ places later moves the survivor $s-1$ places later. This is what we saw for $n=10$ and $s=4$: $J_4(10)=5+3=8$.

Checks. For $s=1$ the formula gives $J_1(n)=1+2\ell=J(n)$, as it should. For $n=10=2^3+2$ we have $\ell=2$, so the survivor is 4 places after s:

s	1	2	3	4	5	6	7	8	9	10
$J_s(10)$	5	6	7	8	9	10	1	2	3	4

From $s=7$ on, $s+4>10$ and we subtract 10: for instance $J_8(10)=8+4-10=2$. We also checked the formula with a computer simulation of the game, for every $n\leq 200$ and every starting person s.

5 Very Large Circles: Base 2

To use the formula $J(n) = 2\ell + 1$, we must first write $n = 2^m + \ell$, that is, find the largest power of 2 not bigger than n. For a large n this is not immediate in base 10. In base 2 it is: the decomposition can be read directly on the digits of n, and the formula becomes a simple operation on these digits. In this whole section, the game starts with person 1, as in Section 4.1: $J(n)$ is the survivor of the usual game.

5.1 Reading 2^m and ℓ in base 2

In base 2, a number is written with the digits 0 and 1, and each digit stands for a power of 2. For instance

$$10 = (1010)_2 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 = 8 + 2.$$

Let $n \geq 1$, with $2^m \leq n < 2^{m+1}$. Then n has exactly $m+1$ digits in base 2, and the first one is a 1:

$$n = (1\, b_{m-1}\, b_{m-2} \cdots b_0)_2 = 2^m + \underbrace{b_{m-1}2^{m-1} + \cdots + b_1 2 + b_0}_{=\ell}.$$

Since each b_i is 0 or 1, the remainder satisfies $0 \leq \ell \leq 2^{m-1} + \cdots + 2 + 1 = 2^m - 1 < 2^m$. So this is exactly the decomposition of Observation 3: the first digit gives 2^m, and the other digits give ℓ. For $n = 10 = (1\,010)_2$ we read $2^3 = 8$ and $\ell = (010)_2 = 2$.

5.2 The rule: move the first digit to the end

Theorem 3 *If the game starts with person* 1 *and* $n = (1\, b_{m-1} \cdots b_1 b_0)_2$, *then*

$$J(n) = (b_{m-1} \cdots b_1 b_0\, 1)_2.$$

In words: to get $J(n)$, *we take the first digit of* n *in base* 2, *which is a* 1, *and move it to the end.*

Proof. By Section 5.1, $\ell = (b_{m-1} \cdots b_0)_2 = \sum_{i=0}^{m-1} b_i 2^i$, and $J(n) = 2\ell + 1$ by Theorem 1. Multiplying by 2 moves every digit one place to the left, and adding 1 writes a 1 in the last place:

$$2\ell + 1 = \sum_{i=0}^{m-1} b_i 2^{i+1} + 1 = b_{m-1}2^m + b_{m-2}2^{m-1} + \cdots + b_0 2^1 + 1 \cdot 2^0 = (b_{m-1} \cdots b_0\, 1)_2.$$

■

Table 2 shows the rule on our sixteen tests. When the digit after the first one is 0, the result starts with 0, and this 0 is simply dropped: for $n = 8 = (1000)_2$ we get $(0001)_2 = 1$.

n	n in base 2	$J(n)$ in base 2	$J(n)$
1	1	1	1
2	10	01	1
3	11	11	3
4	100	001	1
5	101	011	3
6	110	101	5
7	111	111	7
8	1000	0001	1
9	1001	0011	3
10	1010	0101	5
11	1011	0111	7
12	1100	1001	9
13	1101	1011	11
14	1110	1101	13
15	1111	1111	15
16	10000	00001	1

Table 2: The first digit of n moves to the end. The values agree with Table 1.

5.3 What base 2 shows

Base 2 gives another view of Corollary 1.

- $J(n)$ **and** n **have the same number of digits** 1. Moving a digit, or dropping a 0 at the front, does not change the number of 1's. For example $10 = (1010)_2$ and $J(10) = 5 = (101)_2$ both have two.

- $J(n) = 1$ **exactly when** n **is a power of** 2. The result $(b_{m-1} \cdots b_0\, 1)_2$ is equal to 1 only when all the b_i are 0, that is when $n = (10 \cdots 0)_2 = 2^m$.

- $J(n) = n$ **exactly when all the digits of** n **are** 1. The strings $1\, b_{m-1} \cdots b_1 b_0$ and $b_{m-1} \cdots b_1 b_0\, 1$ are equal when $b_{m-1} = 1$, $b_{m-2} = b_{m-1}$, $\ldots$, $b_0 = b_1$, so when all the digits are 1: $n = (11 \cdots 1)_2 = 2^{m+1} - 1$, as for $n = 3$, 7, 15.

5.4 Computing $J(n)$ by hand for a large n

To find $J(n)$ by hand for a large n, we follow four steps:

1. write n in base 2, by dividing by 2 again and again and keeping the remainders;
2. remove the first digit, which is a 1;
3. write a 1 at the end of what is left;
4. convert the result back to base 10.

Example: $n = 1000$**.** We divide by 2 until we reach 0:

number	1000	500	250	125	62	31	15	7	3	1
remainder	0	0	0	1	0	1	1	1	1	1

Reading the remainders from right to left gives $1000 = (1111101000)_2$. We move the first 1 to the end:

$$(1\,111101000)_2 \longrightarrow (111101000\,1)_2 = 512 + 256 + 128 + 64 + 16 + 1 = 977.$$

So $J(1000) = 977$. With the formula: $1000 = 512 + 488$, so $J(1000) = 2 \times 488 + 1 = 977$.

Example: $n = 2026$**.** We have $2026 = (11111101010)_2$. Moving the first digit to the end gives

$$(1\,1111101010)_2 \longrightarrow (1111101010\,1)_2 = 2005,$$

so $J(2026) = 2005$. This agrees with $2026 = 2^{10} + 1002$ and $2 \times 1002 + 1 = 2005$.

Larger values. Table 3 gives a few larger values, all obtained by moving the first digit. We also checked them with the program of Section 6. In the line $n = 10\,000$, two zeros appear at the front after the move and are dropped.

n	n in base 2	$J(n)$	$J(n)$ in base 2
1000	1111101000	977	1111010001
2026	11111101010	2005	11111010101
10 000	10011100010000	3617	111000100001
10^6	11110100001001000000	951 425	11101000010010000001
2^{20}	1 followed by twenty 0	1	1
$2^{20} - 1$	twenty digits 1	1 048 575	twenty digits 1
$2^{20} + 1$	1, nineteen 0, 1	3	11
10^9	111011100110101100101000000000	926 258 177	110111001101011001010000000001

Table 3: Large values of $J(n)$. In each line, n and $J(n)$ have the same number of digits 1.

For $n = 10^6$, there are 20 digits, so $m = 19$, $2^{19} = 524\,288$ and $\ell = 10^6 - 524\,288 = 475\,712$, which gives $J(10^6) = 951\,425$. For $n = 10^9$, the game itself needs 999 999 999 eliminations. With base 2, we only handle the 30 digits of n. In general, n has about $\log_2 n$ digits in base 2, so the work grows like $\log_2 n$ instead of n.

6 The Python Code

We wrote two Python functions during the project. The first one, `J_0`, plays the game step by step. The second one, `J_1`, uses the formula of Section 4. In this section we explain how they work, we compare their cost, and we give a faster version of the formula together with the program we used to check all our results.

6.1 `J_0`: a simulation of the game

Listing 1: The function `J_0`, which simulates the game.

```
def J_0(n):
    if n <= 0:
        return "Please enter a positive integer for n"
    else:
        people = [i for i in range(1, n + 1)]
        print(people)
        while len(people) > 1:
            if len(people) % 2 == 0:  # Even case
                for i in range(len(people)//2):
                    h = people.pop(i + 1)
                    print(people, "\t", h, "is eliminated")
            else:  # Odd case
                for i in range(len(people)//2):
                    h = people.pop(i + 1)
                    print(people, "\t", h, "is eliminated")
                people.insert(0, people.pop(len(people) - 1))
                print(people)
        return f"So {people[0]} is the only survivor"
```

The list. The circle is a Python list, `people`, which starts as $[1, 2, \ldots, n]$. The person at index 0 is the next one to be skipped. If $n \leq 0$ the function returns a message, since the game makes no sense.

One pass over the list. Each turn of the `while` loop goes once through the list and eliminates every second person. The instruction `people.pop(i + 1)` removes the person at index $i + 1$. This index is the right one because the list shrinks while we go: after i eliminations, i people have left the front part of the list, so the person who was at index $2i + 1$ is now at index $i + 1$. For $n = 6$:

$$[1, 2, 3, 4, 5, 6] \xrightarrow{\text{pop}(1)} [1, 3, 4, 5, 6] \xrightarrow{\text{pop}(2)} [1, 3, 5, 6] \xrightarrow{\text{pop}(3)} [1, 3, 5].$$

With $2k$ people the pass makes k eliminations and ends exactly at the end of the list, so the next pass starts again with the person at index 0, who is skipped.

The odd case. With $2k + 1$ people the pass also makes k eliminations, but the last person of the list is skipped at the end of the pass. The next one to be eliminated is therefore the first person of the list. The instruction

```
people.insert(0, people.pop(len(people) - 1))
```

moves the last person to the front. Now he is at index 0 and is skipped, and the person after him is eliminated, as in the game. For instance $[1, 5, 9]$ becomes $[9, 1, 5]$, and the next pass eliminates 1.

Output for $n = 10$. The function prints the list after each elimination:

Listing 2: Output of `J_0(10)`.

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
[1, 3, 4, 5, 6, 7, 8, 9, 10]   2 is eliminated
[1, 3, 5, 6, 7, 8, 9, 10]      4 is eliminated
[1, 3, 5, 7, 8, 9, 10]         6 is eliminated
[1, 3, 5, 7, 9, 10]            8 is eliminated
[1, 3, 5, 7, 9]                10 is eliminated
[1, 5, 7, 9]                   3 is eliminated
[1, 5, 9]                      7 is eliminated
[9, 1, 5]
[9, 5]                         1 is eliminated
[5, 9]
[5]                            9 is eliminated
So 5 is the only survivor
```

The people are eliminated in the order $2, 4, 6, 8, 10, 3, 7, 1, 9$, which is the order of the statement. The passes of the program are not exactly our rounds: in a circle with an odd number of people, we counted the elimination of the first person in the current round (Section 2.1), while the program does it at the start of the next pass. This changes nothing to the order of the eliminations or to the survivor.

6.2 `J_1`: the formula with a slow search

Listing 3: The function `J_1`, which uses the formula $J(n) = 2\ell + 1$.

```python
def J_1(n):
    m = 0
    if n == 1:
        return 1
    elif n <= 0:
        return "Please enter a positive integer for n"
    else:
        # Find the greatest m and the corresponding
        # remainder l such that n = 2^m + l
        for j in reversed(range(n)):
            for i in range(n):
                if n == pow(2, i) + j:
                    m, l = i, j
        return 1 + 2*l
```

Finding m and ℓ. The function tries every pair (i, j) with $0 \leq i, j < n$ and keeps the pairs with $n = 2^i + j$. The outer loop takes j in decreasing order, so the last pair kept has the smallest possible j. A smaller remainder means a larger power of 2, so the last pair kept is the one where 2^i is the largest power of 2 not bigger than n. This is the decomposition of Observation 3, and the function returns $2\ell + 1$.

Example. For $n = 10$, the pairs found are $(0, 9)$, $(1, 8)$, $(2, 6)$ and $(3, 2)$, since $10 = 1 + 9 = 2 + 8 = 4 + 6 = 8 + 2$. The last one is $(3, 2)$, so $m = 3$, $\ell = 2$, and `J_1(10)` returns 5, like `J_0(10)`.

6.3 The cost of `J_0` and `J_1`

`J_0`. The game has $n - 1$ eliminations. Each `pop` may move all the people after the removed one, so one elimination can cost up to n operations, and the whole game

about n^2. On top of that, the function prints the whole list after each elimination, which is also of order n^2 characters. The memory used is of order n, for the list.

`J_1`. The two loops test $n \times n = n^2$ pairs. Moreover, `pow(2, i)` is a number with $i + 1$ digits in base 2, so for large i it is not computed in one step. In practice `J_1` is even slower than `J_0`: in our tests, for $n = 2000$, `J_1` took about 3 seconds, while `J_0` (with the printing switched off) took about a tenth of a second.

So the formula is fast, but this way of using it is not: almost all the work goes into the search for m and ℓ. The next subsection fixes this.

6.4 Faster versions of the formula

To find the largest power of 2 not bigger than n, we do not need to try every pair. We start from $p = 1$ and double p as long as $2p \leq n$. This needs only about $\log_2 n$ steps: 30 steps for $n = 10^9$.

Listing 4: Faster functions for $J(n)$ and for a game that starts with person s.

```
def J_2(n):
    # I look for the largest power of 2 that is not bigger than n,
    # by doubling p as long as I stay below n
    p = 1
    while 2 * p <= n:
        p = 2 * p
    l = n - p
    return 2 * l + 1


def J_binary(n):
    b = bin(n)[2:]              # for example 10 gives '1010'
    return int(b[1:] + b[0], 2)  # I move the first digit to the end


def J_start(n, s):
    # survivor when the game starts with person s (Section 4.2)
    p = 1
    while 2 * p <= n:
        p = 2 * p
    l = n - p
    if s + 2 * l <= n:
        return s + 2 * l
    return s + 2 * l - n
```

The function `J_2` is the formula $J(n) = 2\ell + 1$ with this quick search for 2^m. The function `J_binary` is the rule of Theorem 3: `bin(n)` writes n in base 2 (with the prefix `0b`, which we remove), we move the first digit to the end, and `int(..., 2)` converts back to base 10. The function `J_start` is the formula of Theorem 2. These three functions give the answer for $n = 10^9$, and even for $n = 10^{100}$, at once.

6.5 Checking all the results

To check the formulas, we wrote a second simulation, `play`, which prints nothing and accepts any starting person. It uses a `deque`, a list from which Python can remove the first element quickly.

Listing 5: The simulation used to check every formula.

```python
from collections import deque


def play(n, s=1):
    # I play the game without printing anything, to check the formulas
    circle = deque(range(1, n + 1))
    circle.rotate(-(s - 1))      # person s is now the first of the list
    while len(circle) > 1:
        circle.rotate(-1)        # the first person is skipped and goes to the back
        circle.popleft()         # the next one is eliminated
    return circle[0]


def check(N):
    for n in range(1, N + 1):
        assert play(n) == J_2(n) == J_binary(n)
        for s in range(1, n + 1):
            assert play(n, s) == J_start(n, s)
    return "all the checks passed for n = 1, ..., " + str(N)
```

In `play`, the first person of the deque is always the next one to be skipped. `rotate(-1)` sends him to the back of the circle, and `popleft()` eliminates the person who is now first. The function `check(200)` compares, for every $n \leq 200$, the game with `J_2` and `J_binary`, and for every starting person s, the game with `J_start`. All the checks passed. We also checked that `J_0` and `J_1` give the same survivor as the formula for every $n \leq 300$.

6.6 Comparison of the four functions

Function	Method	Number of steps	$n = 10^9$
`J_0`	plays the game	about n^2	out of reach
`J_1`	formula, search over all pairs	more than n^2	out of reach
`J_2`	formula, doubling search	about $\log_2 n$	at once
`J_binary`	first digit moved to the end	about $\log_2 n$	at once

Table 4: The four functions for $J(n)$.

`J_0` is the one that helped us most at the start: it shows the circle after each elimination, so we could compare it with our drawings. `J_1` was our first use of the formula. `J_2` and `J_binary` show what a formula brings once it is used well: the answer no longer depends on playing the game.

7 Conclusion

We started from a simple game: n people in a circle, every second person eliminated, and the question of who is left at the end.

By playing the game by hand for $n = 1, \ldots, 16$, we saw that person 1 survives when n is a power of 2, and that the survivor moves two places each time a person is added. The key remark was that the game only depends on the places of the people (Lemma 1), so that after one turn of the circle the game is a smaller copy of itself. It gave the two relations

$$J(2k) = 2J(k) - 1, \qquad J(2k+1) = 2J(k) + 1,$$

and, by induction on m, the formula

$$J(2^m + \ell) = 2\ell + 1, \qquad 0 \leq \ell < 2^m,$$

which we proved a second time by induction on n. Since 2^m is the largest power of 2 not bigger than n, it can also be written $J(n) = 2n - 2^{m+1} + 1$ with $m = \lfloor \log_2 n \rfloor$. We then removed the assumption that person 1 starts: if the game starts with person s, the survivor is the person 2ℓ places after s. In base 2, the formula becomes a rule on the digits: the first digit of n moves to the end. This gives the survivor of a circle of 10^9 people at once, where the game would need almost a billion eliminations. Every result was checked by a computer simulation.

The work followed the method of the course: test small cases, look for a pattern, state it precisely, and prove it. Each step was needed. The tests gave the formula, but they could not prove it: sixteen cases say nothing about the seventeenth. The proof, in turn, came from looking closely at one round of the game in the tests. The computer was useful at both ends, first to check our drawings, then to check the formulas on many more cases than we could do by hand. It also showed us that a formula is only fast if it is used well: our first program based on the formula was slower than the simulation.

Some questions remain open for us. The first one is to eliminate every third person instead of every second one, or more generally every q-th person. The proof of Lemma 1 still works: one step now sends the first $q - 1$ people to the back of the list and eliminates the next one. It gives

$$J_q(n) = \big((J_q(n-1) + q - 1) \bmod n\big) + 1,$$

where $a \bmod n$ is the remainder of a divided by n. For $q = 2$ this is relation (1). We checked this relation with a simulation for $q = 3, 4, 5$ and $n < 150$. For $q = 3$ the survivors for $n = 1, \ldots, 16$ are $1, 2, 2, 1, 4, 1, 4, 7, 1, 4, 7, 10, 13, 2, 5, 8$: the survivor goes up by 3 and comes back near the start from time to time, but we did not find a formula as simple as $2\ell + 1$. The case $q = 2$ is special because the circle is exactly halved at each round. A last question remains: can we predict the whole order of the eliminations, and not only the survivor?

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